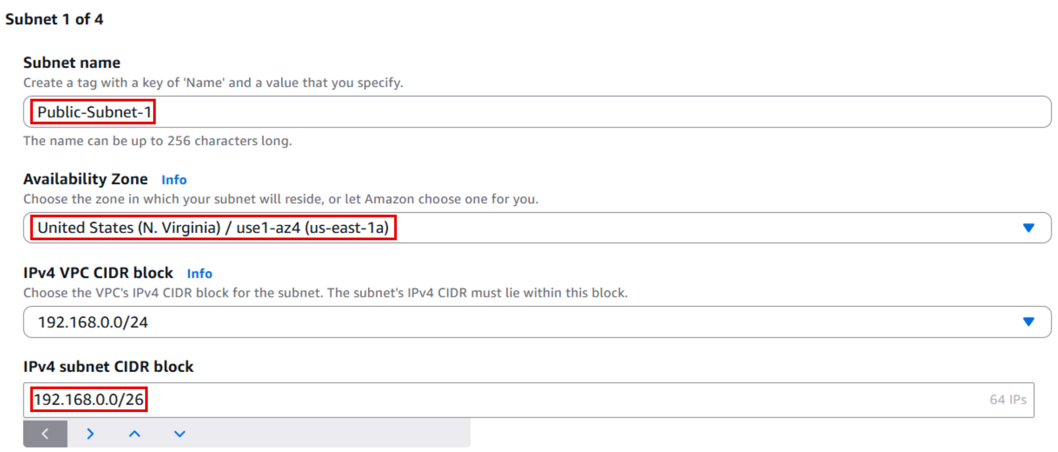
Creating Public and Private Subnets

To Begin with the Lab

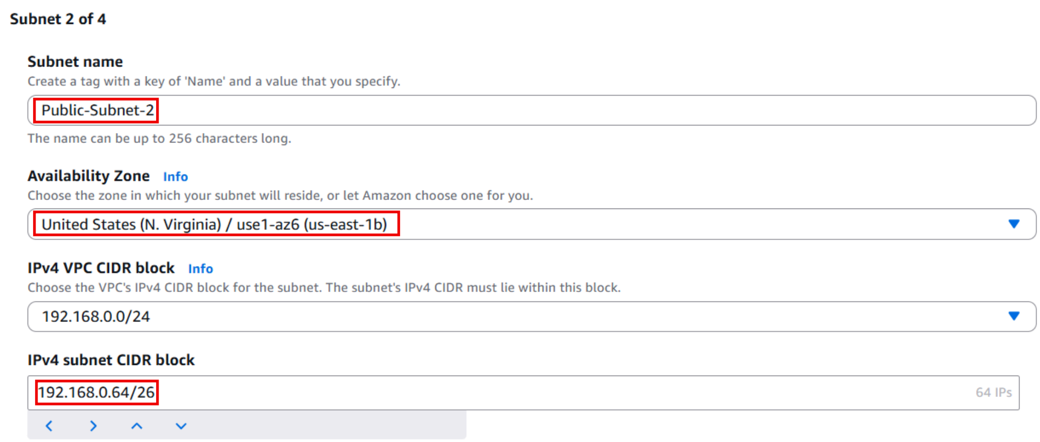
Summary of the Lab

This lab demonstrated how to create a highly available network architecture by configuring two public and two private subnets within an existing custom VPC across different Availability Zones. An Internet Gateway was created and attached to the VPC to enable internet connectivity. A dedicated public route table was then created, associated only with the public subnets, and configured with a default route (0.0.0.0/0) pointing to the Internet Gateway. The private subnets remained associated with the main route table without direct internet access, ensuring secure network isolation while following AWS best practices for scalable and resilient infrastructure.

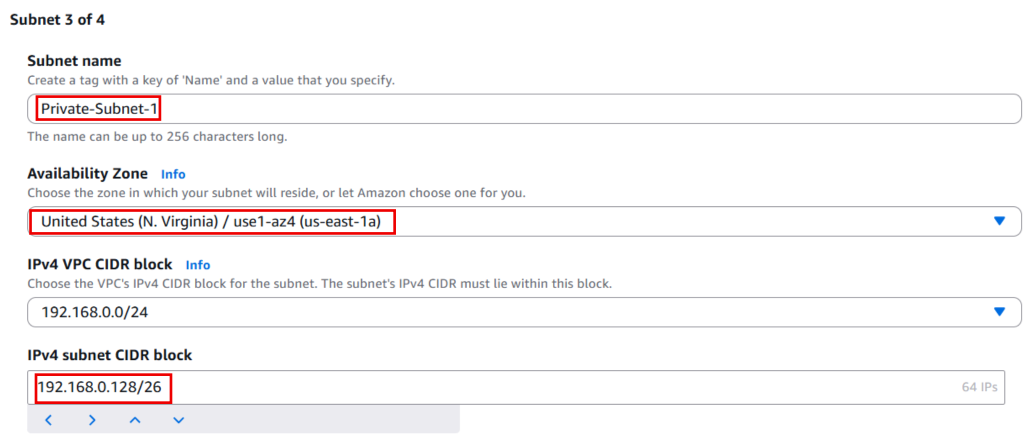
- Note: This lab assumes that the custom VPC (192.168.0.0/24) has already been created. Begin directly with the subnet creation process.
- Please delete the previously created subnets and Internet Gateway as well. We'll recreate them here.
- Open the **AWS Management Console** and navigate to the **VPC** service. Since the VPC is already available, proceed to the **Subnets** section to create separate public and private subnets following AWS best practices.
- Click Create Subnets and select your existing VPC. Instead of creating one subnet at a time, create all four subnets in a single operation so that the VPC is prepared for a highly available application deployment.
- Create the first subnet by naming it Public-Subnet-1, selecting the us-east-1a Availability Zone, and assigning the CIDR block 192.168.0.0/26. This subnet will later be configured as a public subnet.



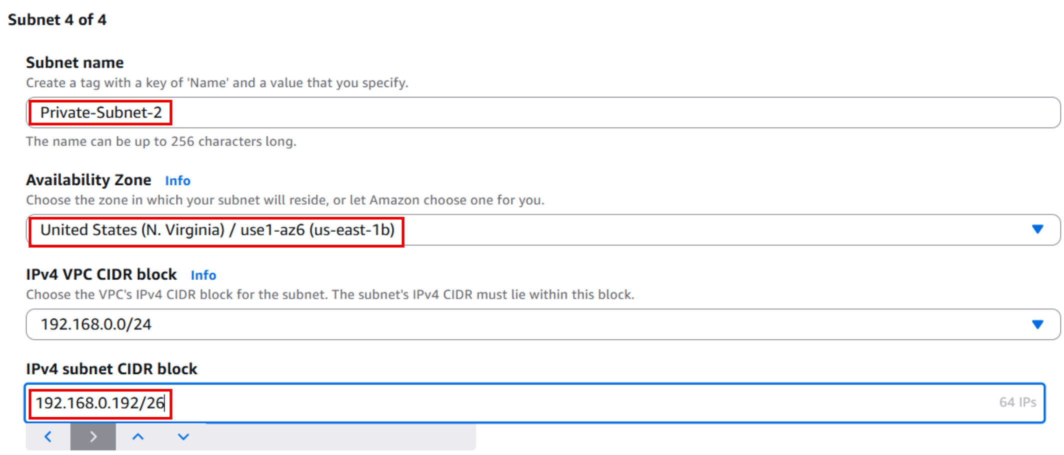
- Add a second subnet named Public-Subnet-2, choose the us-east-1b Availability Zone, and assign the CIDR block 192.168.0.64/26. Placing the subnet in a different Availability Zone improves application availability.



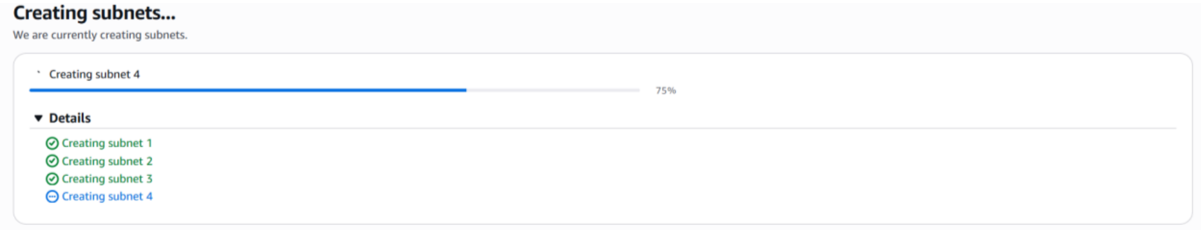
- Add a third subnet named Private-Subnet-1, select the us-east-1a Availability Zone, and assign the CIDR block 192.168.0.128/26. This subnet will later host private resources.



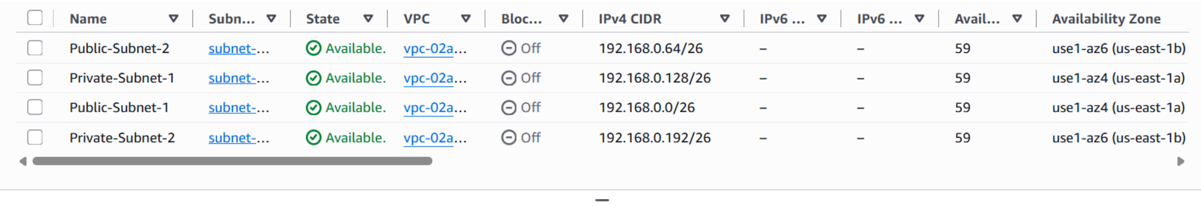
- Create the fourth subnet by naming it Private-Subnet-2, selecting the us-east-1b Availability Zone, and assigning the CIDR block 192.168.0.192/26. After verifying all subnet details



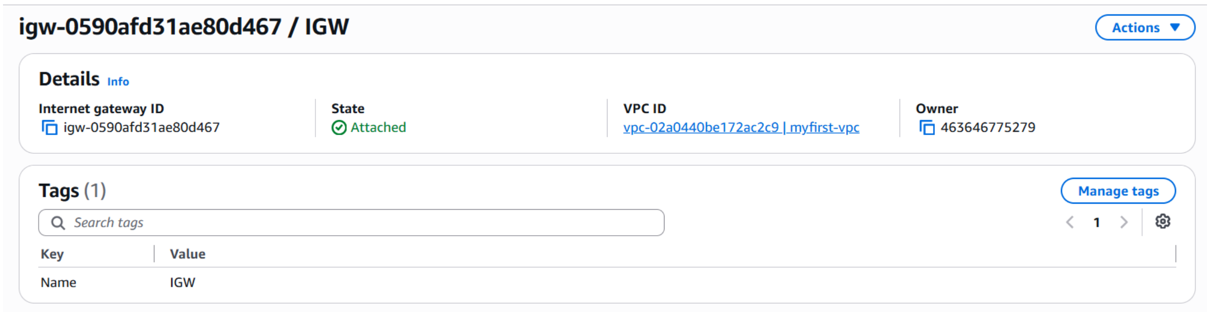
- Click Create Subnets to create all four subnets.



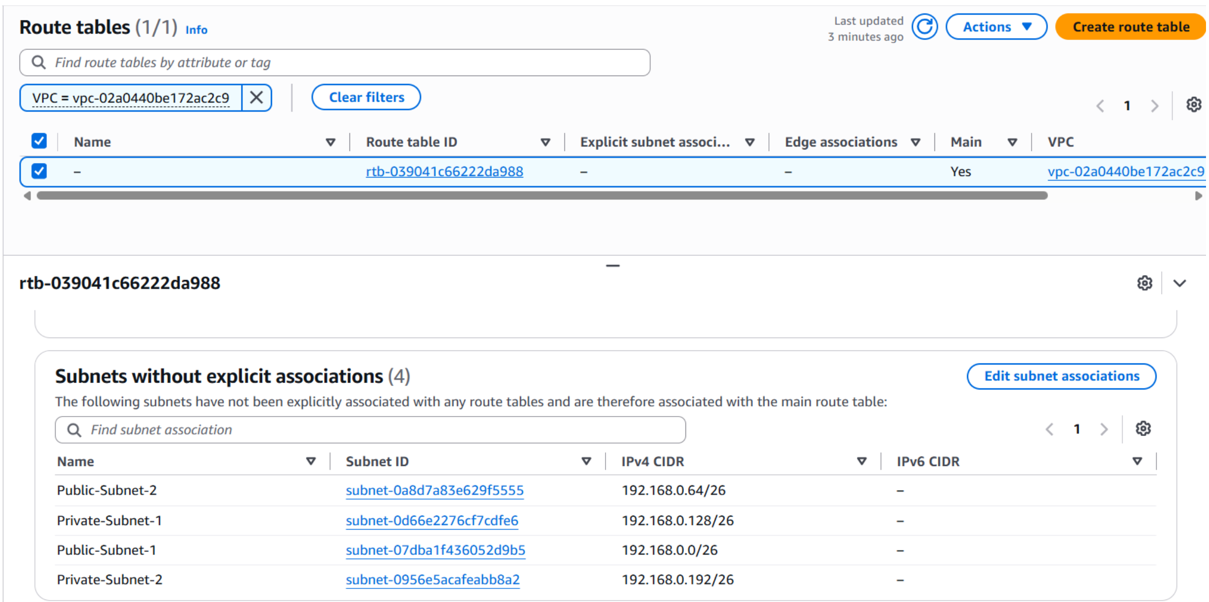
- Four Subnets created Successfully



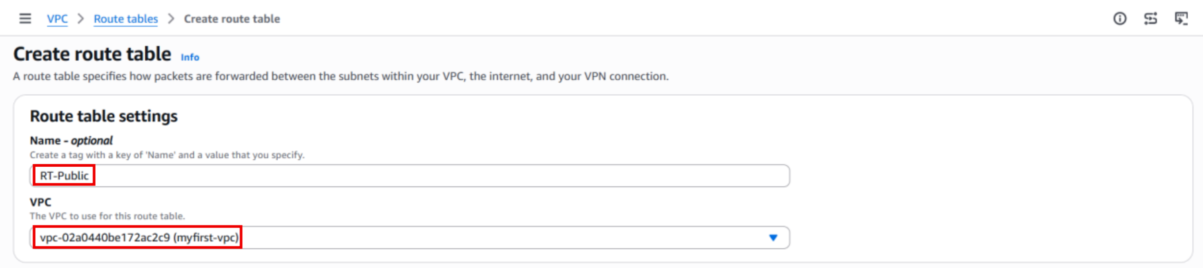
- Navigate to the **Internet Gateways** section and create a new Internet Gateway by providing a suitable name such as **IGW**. After the gateway is created, attach it to the existing VPC so that the VPC can communicate with the internet.



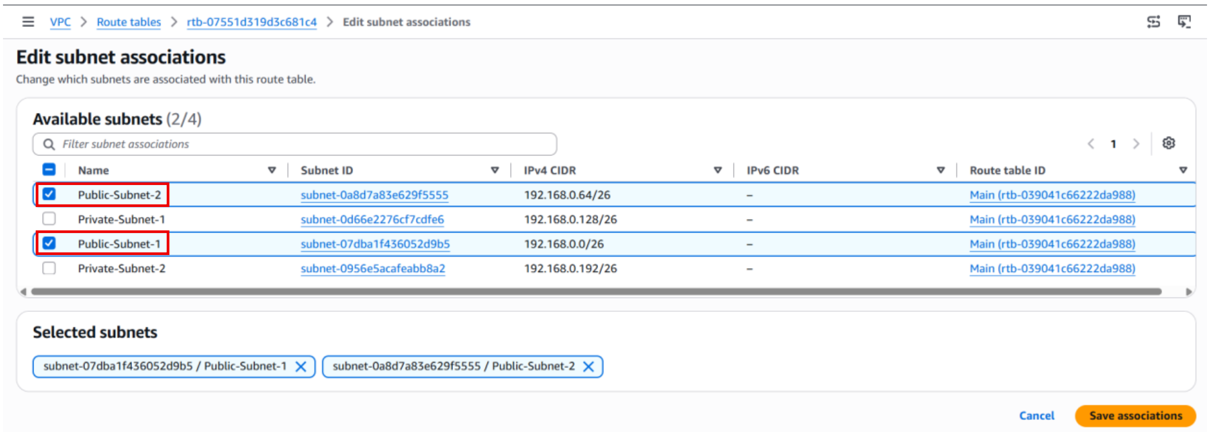
- Open the **Route Tables** section and identify the **Main Route Table** that AWS automatically creates for every VPC. Verify that all four newly created subnets are automatically associated with this route table.



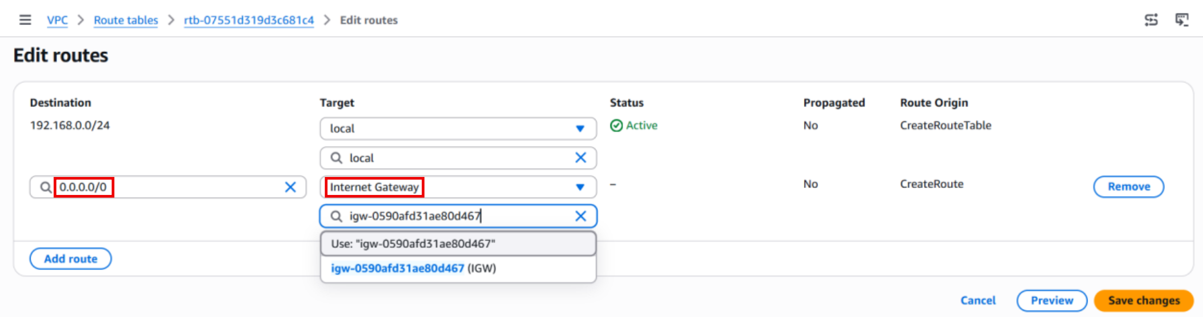
- Create a new Route Table by clicking **Create Route Table**, assigning a name such as **RT-Public**, selecting the existing VPC, and creating the route table. This route table will be used only for the public subnets.



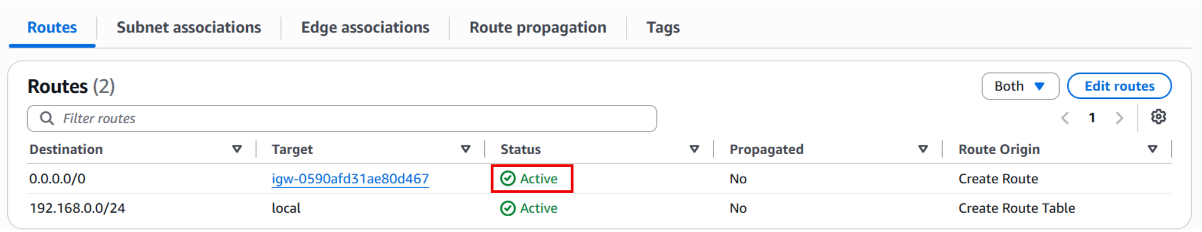
- Open the **Subnet Associations** tab of the newly created **RT-Public** route table and click **Edit Subnet Associations**. Select **Public-Subnet-1** and **Public-Subnet-2**, then save the association so that only the public subnets use this route table.



- Open the **Routes** tab of **RT-Public**, click **Edit Routes**, and add a new route with the destination **0.0.0.0/0**. Select the attached **Internet Gateway** as the target and save the changes to provide internet access for the public subnets.



Verify that the route is in the **Active** state and that the Internet Gateway is attached to the VPC. The public subnets now have internet connectivity, while the private subnets remain associated with the Main Route Table and continue to have no direct internet access.


Routes
Subnet associations
Edge associations
Route propagation
Tags
Routes (2)
Both
Edit routes
Filter routes
1
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0
igw-0590afd31ae80d467
Active
No
Create Route
192.168.0.0/24
local
Active
No
Create Route Table